

Lab Title: Drawing diagrams and budget preparation

Entities, Process, and Data Flows

Process P0: Firearm Information Repository System (FIRS)

Central system for firearm registration, licensing, transactions, compliance, and reporting.

External Entities (Updated) :

1. Customer

- License Application
- License status & Weapon Info.

2. Authorized Dealer

- Sales & Transfer Data
- Weapon Status

3. Govt. Authority

- License Approval
- Monitoring

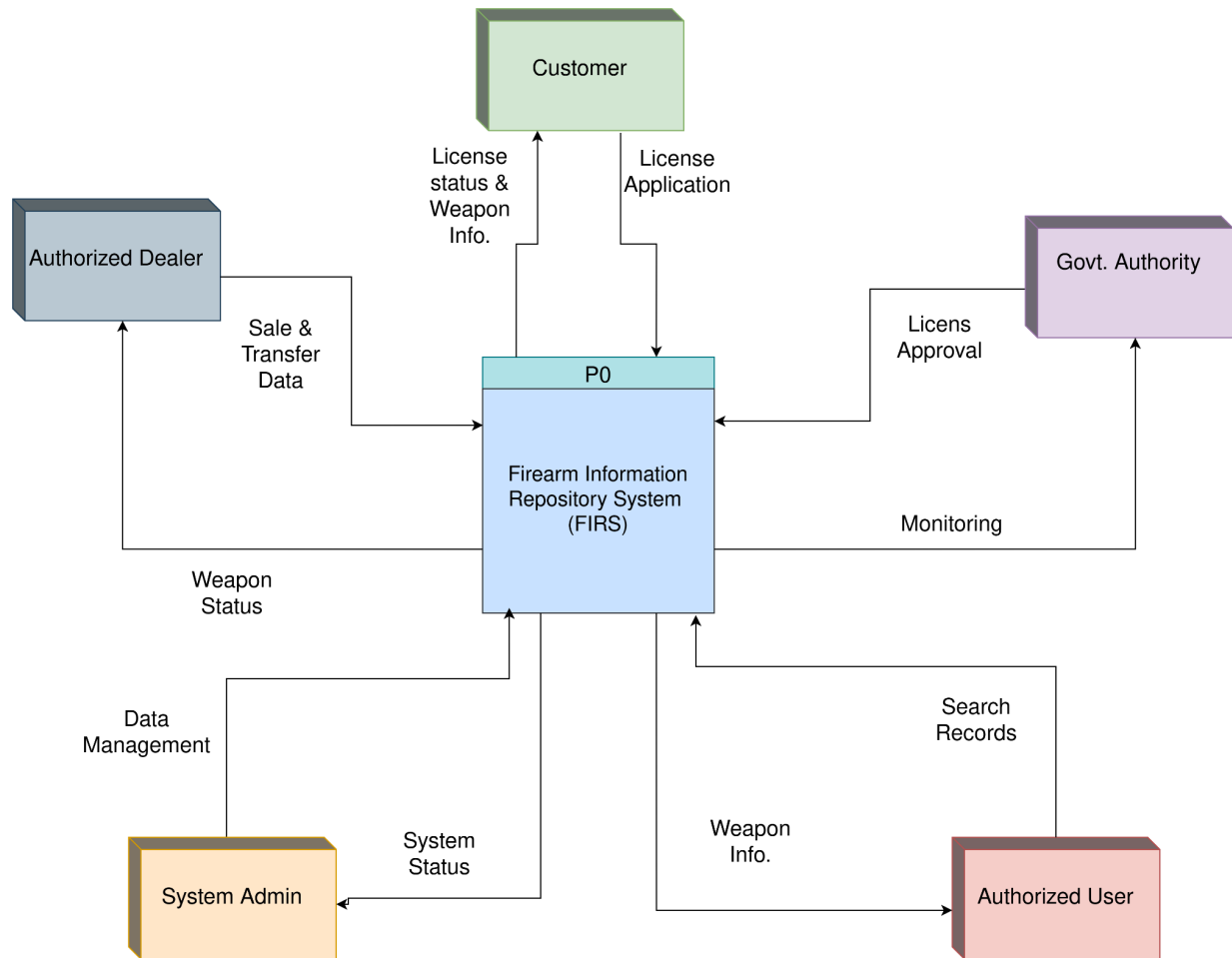
4. Authorized User

- Search Records
- Weapon Info.

5. System Admin

- Data Management
- System Status

The Context level DFD of FIRS



Entities, Types & Relationships

Entities & Attributes

1. Firearm

- Firearm_ID (PK)
- Serial_Number
- Type
- Model
- Status

2. Supplier

- Supplier_ID (PK)
- Name
- Contact_Number

- Address

3. Authorized_User

- User_ID (PK)
- Name
- Rank
- Department

4. Issue_Record

- Issue_ID (PK)
- Issue_Date
- Return_Date
- Firearm_ID (FK)
- User_ID (FK)

5. Administrator

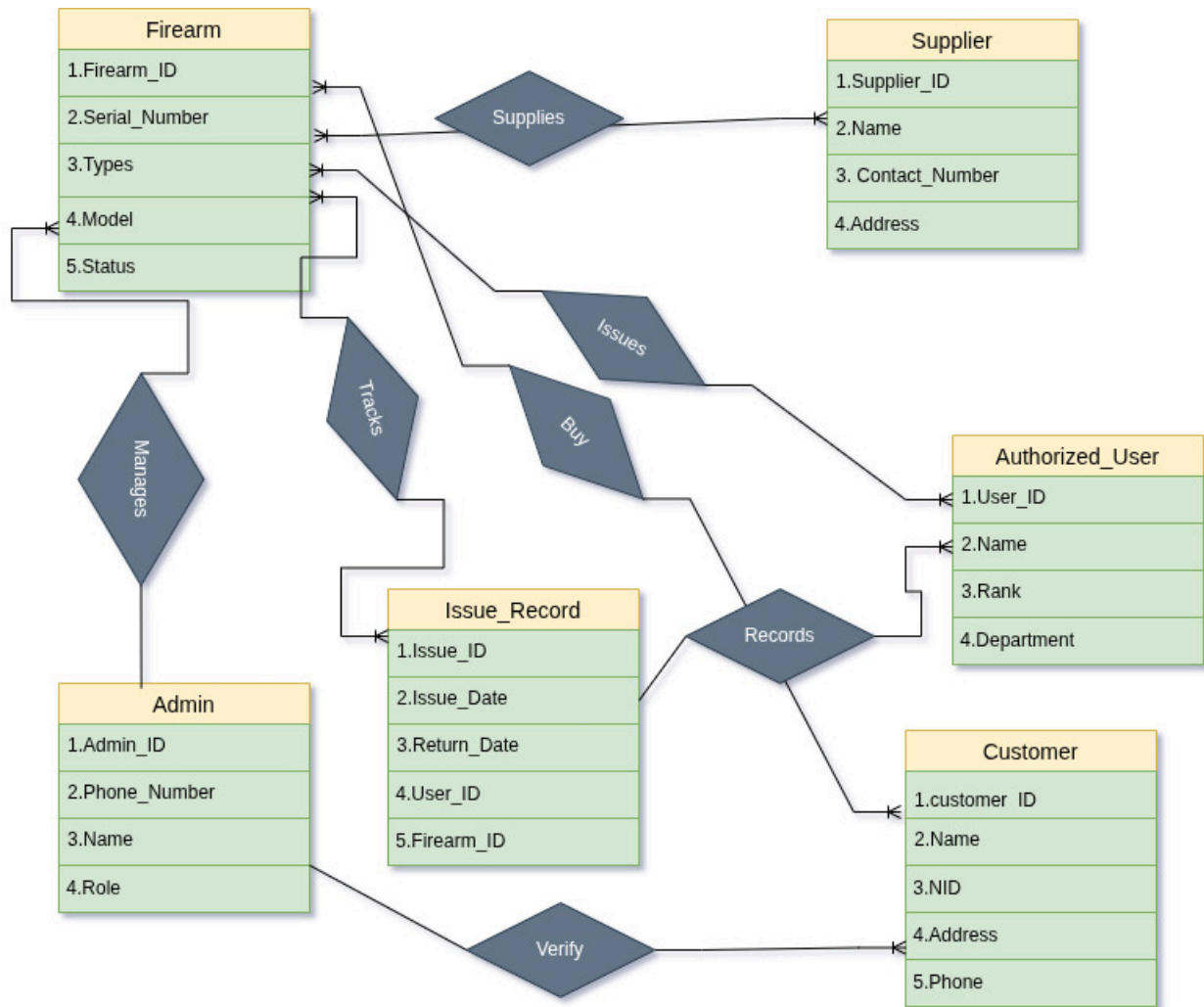
- Admin_ID (PK)
- Name
- Role

6. Customer

- NID(PK)
- Name

- Addres
- Phone

a) The ER –Diagram of FIRS



3. Estimating costs breakdown and preparing the budget (Total)

PROJECT TITLE				Firearm Information Repository System (FIRS)			
PROJECT MANAGER				Abu Ahmad Mahfuzor Rahman			
WBS NUMBER	TASK TITLE	TASK OWNERS	HOURS		AVG RATE	COST	
			MIN	MAX		MIN	MAX
1	Concept development	Abu Ahmed Mahfuzur Rahman, Md Fahim Islam , Md. Mahfuzul Bashar Shaikot	20	30	40	800	1200
2	Planning	Md Fahim Islam, Md. Mahfuzul Bashar Shaikot	30	34	40	1200	1360
3	Requirements analysis	Md Fahim Islam, Md. Mahfuzul Bashar Shaikot	20	28	40	800	1120
4	Design	Al Adid, Md Fahim Islam, Md. Mahfuzul Bashar Shaikot, Abu Ahmad Mahfuzor Rahman, Md. Sakibur Rahman	50	60	30	1500	1800
5	Development	Md Fahim Islam, Md. Mahfuzul Bashar Shaikot	2200	2400	30	66000	72000
5.1	Deliverable #1	Md Fahim Islam, Md. Mahfuzul Bashar Shaikot	1400	1500	35	49000	52500
5.2	Deliverable #2	Md Fahim Islam, Md. Mahfuzul Bashar Shaikot,Md. Sakibur	1300	1400	37	48100	51800

		Rahman					
5.3	Deliverable #3	Md Fahim Islam, Md. Mahfuzul Bashar Shaikot, Md. Sakibur Rahman	1600	1700	30	48000	51000
5.4	Deliverable #4	Md Fahim Islam, Md. Mahfuzul Bashar Shaikot, Md. Sakibur Rahman	1800	1900	37	66600	70300
5.5\	Deliverable #5	Md Fahim Islam, Md. Mahfuzul Bashar Shaikot, Md. Sakibur Rahman	80	100	40	3200	4000
6	Reporting and Mesurement	Md Fahim Islam, Md Mahfuzul Bashar Shaikot	200	240	35	7000	8400
7	Implementation	Abu Ahmad Mahfuzor Rahman, Al Adid	50	60	25	1250	1500
8	Quality assurance	Abu Ahmad Mahfuzor Rahman, Al Adid, Md Fahim Islam, Md Mahfuzul Bashar Shaikot, Md Sakibur Rahman	400	500	28	11200	14000
9	QA Testing	Abu Ahmad Mahfuzor Rahman, Al Adid	600	700	33	19800	23100

10	Maintainance	Md Sakibur Rahman, Md Fahim Islam, Md Mahfuzul Bashar Shaikot	140	150	35	4900	5250
11	Launch	Al Adid, Md Sakibur Rahman, Md Fahim Islam, Md Mahfuzul Bashar Shaikot, Abu Ahmad Mahfuzor Rahman	130	135	140	12000	14000
12	Close down	Md Fahim Islam, Md. Mahfuzul Bashar Shaikot, Md. Sakibur Rahman	35	55	600	3450	13000
Totals			10055	10992	1255	344800	386330